

IMPACT ANALYSIS

CR-012: Dark Mode and Light Mode for the Full System

1. Change Summary

CR-012 introduces Dark Mode and Light Mode throughout the Student Management System. A theme toggle will be available on applicable pages. Theme preferences are section-specific: Login, Admin and Student themes are maintained separately. Admin child pages follow the Admin theme, while Student child pages follow the Student theme.

2. Impact Assessment

Area	Impact	Assessment
Frontend	Low - Moderate	Theme toggle, theme state and CSS styling will be added or updated.
Backend	None	No backend business logic changes are required.
API	None	Existing APIs remain unchanged.
Database	None	No schema or student data changes are required.
Authentication	None	Existing authentication remains unchanged.
Authorization	None	Existing Admin/Student access controls remain unchanged.
Performance	Low	Theme switching is handled on the client side.
CRUD	None	Existing Add/Edit/Delete/View/Search operations remain unchanged.
Deployment	Low	No database migration or new backend endpoint is required.

3. Frontend Impact

- Dark/Light theme toggle on applicable pages.
- Separate Login, Admin and Student theme preferences.
- Admin pages use the selected Admin theme.
- Student pages use the selected Student theme.
- CSS variables and theme-specific styles for UI elements.
- Theme persistence using localStorage.
- Responsive styling for both themes.
- Existing page functionality remains unchanged.

4. Backend, API and Database Impact

No backend changes are expected.

- No new API endpoint is required.
- Existing API requests remain unchanged.
- No database schema changes are required.
- No student records or academic data are modified.
- Theme preference is stored client-side using localStorage.

5. Security Impact

Existing authentication and authorization mechanisms remain unchanged.

- Must not bypass authentication or authorization.
- Must not expose additional sensitive information.
- Theme storage contains only a non-sensitive preference.
- Existing backend security controls remain unchanged.
- No passwords, tokens or authentication data are stored for the theme feature.

6. CRUD and Functional Impact

CR-012 is a visual enhancement and must not change existing CRUD or functional behaviour.

- Add Student continues to work.
- Edit Student continues to work.
- Delete Student continues to work.
- Search and filtering continue to work.
- Student records remain unchanged when switching themes.
- Theme changes affect presentation and preference only.

7. Performance Impact

Expected performance impact is low because theme switching is performed on the client side. No additional API requests or database operations are required.

- No API request for every theme toggle.
- No database query for theme switching.
- Client-side state and CSS provide fast changes.
- No unnecessary page reloads.

8. Testing Impact

- Login Dark/Light Mode.
- Admin Dark/Light Mode.
- Student Dark/Light Mode.

- Admin and Student themes remain separate.
- Login theme remains separate.
- Admin Add/Edit pages use Admin theme.
- Student Profile use Student theme.
- Persistence after navigation and refresh..
- Existing Login/Logout and Add/Edit/Delete/Search.
- Authorization and protected pages.
- No API/database regression.
- No console/runtime errors.

9. Risk Analysis

Risk	Impact	Mitigation
Theme changes another section	High	Use separate Login, Admin and Student theme state/keys.
UI unreadable in Dark Mode	Medium	Test all major UI elements in both themes.
Existing CRUD breaks	High	Perform regression testing for Add/Edit/Delete/Search.
Authentication/authorization affected	High	Keep security logic unchanged and test protected pages.
Theme lost after refresh	Medium	Use and test localStorage persistence.
Unnecessary API/database requests	Low	Keep theme handling client-side.
Responsive UI problems	Low	Test supported screen sizes.

10. Overall Impact and Recommendation

Overall Impact: Low

CR-012 is primarily a frontend visual and usability enhancement. It does not change backend logic, APIs, database structure, student data, authentication, authorization or CRUD functionality.

The change is technically feasible and recommended for implementation, subject to formal approval and testing. The key consideration is maintaining separate Login, Admin and Student theme preferences while preserving all existing functionality.